

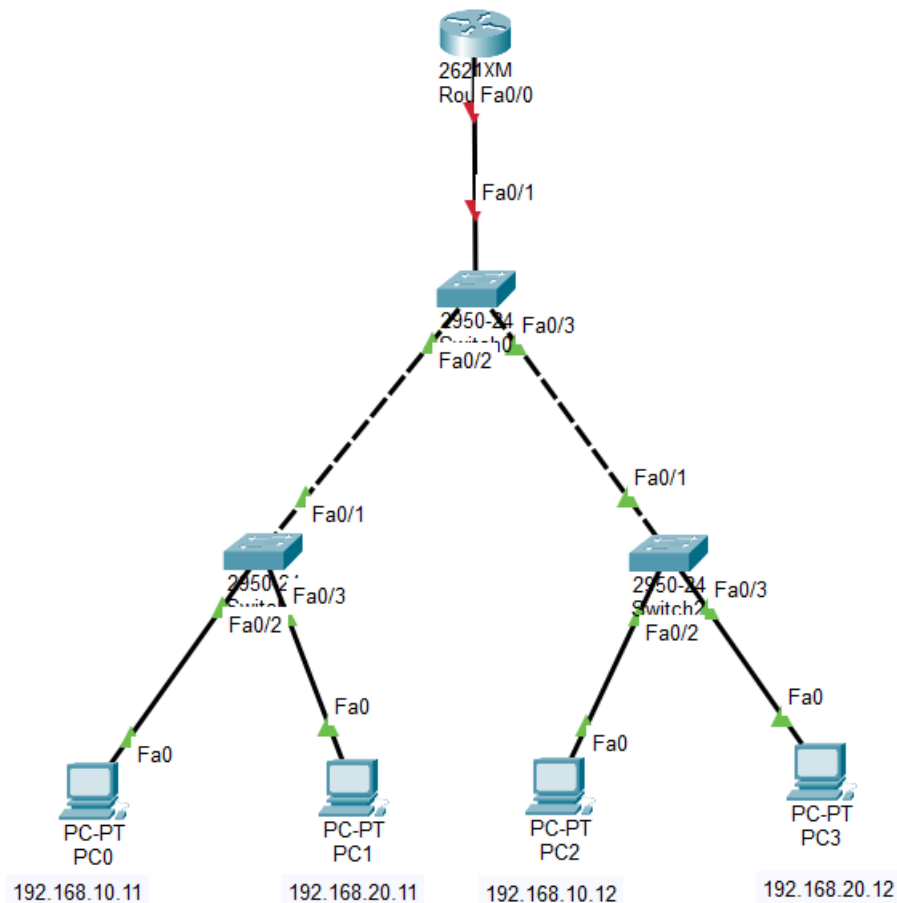
Weekly Review and Practice Practical (2 hours): Create a network with multiple VLANs and configure inter-VLAN routing. Task: Submit a report on VLAN and inter-VLAN routing setup.

Task 1.

On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.

Device	PC	VLAN	Department
S1	PC1	10	Sales
S1	PC2	20	Support
S2	PC3	10	Sales
S2	PC4	20	Support

PC	IP address	Gateway	VLAN
PC1	192.168.10.11	192.168.10.1	10
PC2	192.168.20.11	192.168.20.1	20
PC3	192.168.10.12	192.168.10.1	10
PC4	192.168.20.12	192.168.20.1	20



Task 2.

Configure trunk ports between the switches to allow VLAN traffic to pass through. Verify trunking by using the show interfaces trunk command and submit a screenshot of the output.

```

Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      1
Fa0/2     on        802.1q         trunking      1
Fa0/3     on        802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1     1-1005
Fa0/2     1-1005
Fa0/3     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     1,10,20
Fa0/2     1,10,20
Fa0/3     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     1,10,20
Fa0/2     1,10,20
Fa0/3     1,10,20

Switch#

```

Task 3.

Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter-VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.

```
Router#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Router#show ip interface brief
Interface                IP-Address      OK? Method Status              Protocol
FastEthernet0/0          unassigned      YES unset    up                  up
FastEthernet0/0.10       192.168.10.1    YES manual    up                  up
FastEthernet0/0.20       192.168.20.1    YES manual    up                  up
FastEthernet0/1          unassigned      YES unset    administratively down down
Router#
```

PC	VLAN	IP Address	Default Gateway
PC0	10	192.168.10.11	192.168.10.1
PC1	20	192.168.20.11	192.168.20.1
PC2	10	192.168.10.12	192.168.10.1
PC3	20	192.168.20.12	192.168.20.1

Task 4.

Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works.

Hint: Check sub-interface IPs, trunk configuration, and default gateways on PCs.

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Reply from 192.168.20.11: bytes=32 time=1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```